

Renato Paredes Venero

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PROFILE

Computational Neuroscientist. I study the neural basis of cognition and behaviour in neurological and psychiatric conditions through experimental research and computational modelling. My main interest is multisensory integration processes underlying bodily self alterations. I also actively collaborate in the field of human-robot interaction, focusing on the validation of emerging technologies for mental health and education. I currently work as an Assistant Professor at Pontifical Catholic University of Peru, where I lead the Computational Cognitive Neuroscience Lab.

RESEARCH EXPERIENCE

Pontifical Catholic University of Peru

Head of Computational Cognitive Neuroscience Lab

Jan 2023 – Present

- I started the implementation of this new lab in the Department of Psychology, actively setting standards for workflow, equipment, human resources and software management.
- I lead a team of 16 people among researchers, assistants, and students working on cognitive neuroscience projects using computational methods (More details: laboratorio-dptopsicologia.pucp.edu.pe).

Principal Investigator (Department of Psychology)

Aug 2023 – Present

- Project: "Towards the development of a neurocomputational profile of Peruvian women with symptoms of CPTSD"
- Project: "Multisensory integration and auditory sensitivity in students with high abilities at a private university in Metropolitan Lima"
- Project: "Melodies of the mind: Deciphering the impact of music on the regulation of affective and cognitive states of university students using EEG and NLP"

Co-researcher (Department of Engineering)

Apr 2021 – Apr 2023

- Project: "Teleoperated Mobile Robot for Mental Health Delivery in Patients with Infectious Diseases"

Research Assistant (Department of Psychology)

Mar 2016 – May 2018

- Project: "Neurocognitive study of perception of actions and emotions in the interaction between human beings and humanoid robots"

Junior Researcher (DGI)

Set 2015 – May 2016

- Project: "Implementation of a non-invasive EEG System for the study of Human Mirror Neurons".

Max Planck Research Unit for Neurogenetics

Doctoral Researcher

Dec 2024 – Apr 2025

- I conducted computational and statistical analysis of STPT images of the mouse olfactory bulb.

The University of Edinburgh

Remote Research Assistant (Computational Psychiatry Lab)

Sep 2019 – Aug 2021

- I started the collaboration with two laboratories of world-class universities (NYU & University of Chieti-Pescara).
- I led the process of writing and publishing two articles on neural networks in high impact journals.

University of Waseda

Data Analysis Consultant (School of International Liberal Studies)

May – Sep 2018

- I implemented Generalized Linear Models for data analysis in Human-Robot Interaction (HRI) studies.
- I co-authored papers in relevant HRI journals and conferences.

TEACHING EXPERIENCE

Pontifical Catholic University of Peru

Assistant Professor (Department of Psychology)

Jan 2025 – Present

- Neuropsychology

Full-time Lecturer (Department of Psychology)

Aug 2022 – Dec 2024

Undergraduate

- Behavioural Neuroscience
- Neuropsychology
- Introduction to Python for Psychological Research
- Non experimental research methods and Statistics

Postgraduate

- Statistics and Mathematical Psychology
- Research Methods for Psychology

Adjunct Lecturer (Department of Psychology)

Jan 2021 – July 2022

Teaching Assistant (Department of Psychology)

Mar 2017 – Sep 2018

- Research Methods and Statistics I

Continuing Education Teacher (CETAM)

Jan – Apr 2018

- Research Methods for Human-Machine Interaction

Neuromatch Academy Inc.

Teaching Assistant

Jun 2020 – Jul 2022

- Support students in learning the theory and techniques of deep learning with an emphasis on neuroscience.
- Support students in learning computational neuroscience and analytical modelling of brain activity.
- Review of didactic material for theoretical and practical learning sessions.

EDUCATION

National University of Cordoba

Cordoba, Argentina

PhD in Neuroscience (to be awarded)

Jun 2021 - Apr 2025

Supervisors: Dr. Peggy Series and Dr. Pablo Barttfeld

Topic: "Neural Network modelling of multisensory integration in patients with schizophrenia and autism spectrum disorder"

Defence date: 09/04/2025

The University of Edinburgh

Edinburgh, UK

MSc. in Cognitive Science, with Distinction and Dissertation Prize

Sep 2018 - Aug 2019

Supervisor: Dr. Peggy Series

Topic: "Computational modelling of peripersonal space representation in schizophrenia spectrum disorders"

Pontifical Catholic University of Peru

Lima, Peru

B.A. in Educational Psychology, with Outstanding Mention

Mar 2010 - Jul 2016

AWARDS

- Winner of the Annual Research Project Grant (CAP - PUCP 2023-2024).
- Winner of the Research Support Grant (FAI - PUCP 2023).
- Cognitive Science Msc Dissertation Prize (The University of Edinburgh)
- First position of the Excellence Scholarship "President of the Republic" - 2018 granted by PRONABEC (MINEDU).
- Best paper award in the category "Excellence in Engineering Education" at IEEE Global Engineering Education Conference (EDUCON 2017).
- Thesis approved unanimously with outstanding mention during undergraduate studies.
- Winner of the Programme to Support Initiation in Research (PAIN - PUCP 2015).
- Scholarship holder of the student mobility program of the Interuniversity Development Centre (CINDA 2013).

PARTICIPATION IN EVENTS

- Scientific Committee at X Congreso Latinoamericano de Psicología de la Salud (ALAPSA).
- Oral presentation at Adriatica Summer School 2024.
- Oral presentation at the ALADAA International Conference 2023.
- Poster presentation at Bernstein Conference 2021.
- Reviewer at international conferences on Human-Robot Interaction: ICSR (2020) and RO-MAN (2017-2024).
- Programme Committee and paper presentation at the International Workshop on Evaluation Methods Standardisation for Human-Robot Interaction (EMSHRI 2017).

SEMINARS AND TALKS

- MAR 2025: ANC Workshop at Institute for Adaptive and Neural Computation (The University of Edinburgh)
- FEB 2025: Lab Meeting at Laboratory of Cognitive Neuroscience (EPFL Campus Biotech, Geneva)
- APR 2025: Lab Meeting at Multisensory Processing Lab (UCLA, Los Angeles)

INFORMATICS SKILLS

- Programming Languages: Python, R, SQL, Java
- Scientific software development: Python scientific stack, Pytest, Tox
- Technical documentation: LaTeX, Overleaf, Markdown, Sphinx
- Database management: MySQL, Hive, Impala
- Development environments: Linux, RStudio, Visual Studio Code, Google Colab, Git

PUBLICATIONS

Journal Articles

- Paredes, R.,** Cabral, J. B., & Seriès, P. (2025a). Scikit-NeuroMSI: A Generalized Framework for Modeling Multisensory Integration. *Neuroinformatics*, 23(3), 40. <https://doi.org/10.1007/s12021-025-09738-1>
- Paredes, R.,** Ferri, F., Romei, V., & Seriès, P. (2025b). Increased excitation enhances the sound-induced flash illusion by impairing multisensory causal inference in the schizophrenia spectrum. *Schizophrenia Research*, 283, 1–10. <https://doi.org/10.1016/j.schres.2025.06.007>
- Pérez-Zuñiga, G., Arce, D., Gibaja, S., Alvites, M., Cano, C., Bustamante, M., Horna, I., **Paredes, R.,** & Cuellar, F. (2024). Qhali: A Humanoid Robot for Assisting in Mental Health Treatment. *Sensors*, 24(4), 1321.
- Gomez-Quispe, J. M., Pérez-Zuñiga, G., Arce, D., Urbina, F., Gibaja, S., **Paredes, R.,** & Cuellar, F. (2023). Non Linear Control System for Humanoid Robot to Perform Body Language Movements. *Sensors*, 23(1), 552.

- Paredes, R., Ferri, F., & Seriès, P.** (2022). Influence of E/I balance and pruning in peri-personal space differences in schizophrenia: a computational approach. *Schizophrenia Research*, 248, 368–377. <https://doi.org/10.1016/j.schres.2021.06.026>
- Noel, J.-P., **Paredes, R.**, Terrebonne, E., Feldman, J. I., Woynaroski, T., Cascio, C. J., Seriès, P., & Wallace, M. T. (2022). Inflexible updating of the self-other divide during a social context in autism: psychophysical, electrophysiological, and neural network modeling evidence. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, 7(8), 756–764. <https://doi.org/10.1016/j.bpsc.2021.03.013>
- Marius’t Hart, B., Achakulvisut, T., Adeyemi, A., Akrami, A., Alicea, B., Alonso-Andres, A., Alzate-Correa, D., Ash, A., Ballesteros, J. J., Balwani, A., et al. (2022). Neuromatch Academy: a 3-week, online summer school in computational neuroscience. *Journal of Open Source Education*, 5(49), 118. <https://doi.org/10.21105/jose.00118>
- Trovato, G., De Saint Chamas, L., Nishimura, M., **Paredes, R.**, Lucho, C., Huerta-Mercado, A., & Cuellar, F. (2021). Religion and Robots: Towards the Synthesis of Two Extremes. *International Journal of Social Robotics*, 13, 539–556. <https://doi.org/10.1007/s12369-019-00553-8>
- Trovato, G., Lopez, A., **Paredes, R.**, Quiroz, D., & Cuellar, F. (2019). Design and Development of a Security and Guidance Robot for Employment in a Mall. *International Journal of Humanoid Robotics*, 16(5). <https://doi.org/10.1142/S0219843619500270>
- Trovato, G., Lucho, C., & **Paredes, R.** (2018a). She’s Electric—The Influence of Body Proportions on Perceived Gender of Robots across Cultures. *Robotics*, 7(3). <https://doi.org/10.3390/robotics7030050>

Conference Papers

- Paredes, R., Cabral, J., & Seriès, P.** Scikit-NeuroMSI: a Python framework for multisensory integration modelling. In: *IX Congreso de Matemática Aplicada, Computacional e Industrial*. 9. 2023, 545–548.
- Arce, D., Gibaja, S., Urbina, F., Maura, C., Huanca, D., **Paredes, R.**, Cuellar, F., & Pérez-Zuniga, G. Design and preliminary validation of social assistive humanoid robot with gesture expression features for mental health treatment of isolated patients in hospitals. In: *Social Robotics: 14th International Conference, ICSR 2022, Florence, Italy, December 13–16, 2022, Proceedings, Part II*. Springer. 2023, 518–528.
- Paredes, R., & Seriès, P.** Influence of E/I balance in peri-personal space differences in schizophrenia and autism: a neural network approach. In: *Bernstein Conference 2021*. 2021. <https://doi.org/10.12751/nnn.bc2021.p028>.
- Trovato, G., **Paredes, R.**, Balvin, J., Cuellar, F., Thomsen, N. B., Bech, S., & Tan, Z.-H. The Sound or Silence: Investigating the Influence of Robot Noise on Proxemics. In: *2018 27th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*. 2018, 713–718. <https://doi.org/10.1109/ROMAN.2018.8525795>.
- Abarca, M., Saito, C., Cerna, J., **Paredes, R.**, & Cuéllar, F. An interdisciplinary unmanned aerial vehicles course with practical applications. In: *2017 IEEE Global Engineering Education Conference (EDUCON)*. 2017, 255–261. <https://doi.org/10.1109/EDUCON.2017.7942856>.
- Lopez, A., Ccasane, B., **Paredes, R.**, & Cuellar, F. Effects of using indirect language by a robot to change human attitudes. In: *Proceedings of the companion of the 2017 ACM/IEEE international conference on human-robot interaction*. 2017, 193–194. <https://doi.org/10.1145/3029798.3038310>.
- Paredes, R.**, Laurel, C., Cuellar, F., & Davila, A. Implementation of Synchronization Triggers in an open-source EEG system for Visual Evoked Potentials measurement. In: *SAN2016 Meeting*. 2016. <https://doi.org/10.3389/conf.fnhum.2016.220.00007>.
- Caceres, P. C., **Venero, R. P.**, & Cordova, F. C. Tangible programming mechatronic interface for basic induction in programming. In: *2018 IEEE Global Engineering Education Conference (EDUCON)*. 2018, 183–190. <https://doi.org/10.1109/EDUCON.2018.8363226>.

- Trovato, G., Lopez, A., **Paredes**, R., & Cuellar, F. Security and guidance: Two roles for a humanoid robot in an interaction experiment. In: *2017 26th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*. 2017, 230–235. <https://doi.org/10.1109/ROMAN.2017.8172307>.
- Lopez, A., **Paredes**, R., Quiroz, D., Trovato, G., & Cuellar, F. Robotman: A security robot for human-robot interaction. In: *2017 18th International Conference on Advanced Robotics (ICAR)*. 2017, 7–12. <https://doi.org/10.1109/ICAR.2017.8023489>.

Book Chapters

- Paredes Venero**, R., & Davila, A. (2020). Experimental Research Methodology and Statistics Insights. In C. Jost, B. Le Pévédic, T. Belpaeme, C. Bethel, D. Chrysostomou, N. Crook, M. Grandgeorge, & N. Mirnig (Eds.), *Human-Robot Interaction: Evaluation Methods and Their Standardization* (pp. 333–353). Springer, Cham. https://doi.org/10.1007/978-3-030-42307-0_13